

Spring 2012

Problem 1 (Sp12, Sp21). Suppose that X is a compact metric space. If Y is another metric space (possibly noncompact), let $\pi : X \times Y \rightarrow Y$ be the map $\pi(x, y) = y$. Show that if Z is a closed subset of $X \times Y$ then $\pi(Z)$ is closed in Y .

Problem 2 (Su80, Sp12, Sp21, Sp22, Fa22). Let G be a finite group and H be a subgroup.

1. Show that the number of subgroups of G conjugate to H divides the index of H .
2. Show that if $G = \bigcup_{g \in G} gHg^{-1}$ then $G = H$.

Problem 3 (Sp12). The *moments* of a function f are the numbers

$$\int_0^\infty x^n f(x) dx \quad \text{for } n = 0, 1, 2, \dots$$

Find the moments of $f(x) = \exp(-x^{1/4}) \sin(x^{1/4})$.

Problem 4 (Sp12, Sp21). Find the eigenvalues of the $n \times n$ matrix A with entries a_{ij} , where

$$a_{ij} = \begin{cases} 1 & i = j + 1 \\ -1 & i = j - 1 \\ 0 & \text{otherwise} \end{cases}$$

Problem 5 (Sp12). Find the solution of the differential equation

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 0$$

in $x > 0$, such that $y(1) = 0$ and $\frac{dy}{dx} = 1$ at $x = 1$.

Problem 6 (Sp12). A positive integer m is called a *pseudoprime* to the base 2 if m divides $2^{m-1} - 1$. Show $2^p - 1$ is a pseudoprime to the base 2 if p is prime.

Problem 7 (Sp12). Show that the unit circle is a natural boundary of the function $f(z) = \sum_{n \geq 0} z^{2^n}$; in other words, $f(z)$ cannot be extended to an analytic function on any connected open set strictly larger than the open unit disc.

Problem 8 (Sp12, Sp21). Suppose L is a linear operator acting on a non trivial vector space V over a field $\mathbb{K}$. Suppose $p(x) \in \mathbb{K}[x]$ is not identically zero and $p(L) = 0$. Show every eigenvalue of L is a root of p . Show that if p factors completely over $\mathbb{K}$ then some roots of p are eigenvalues of L .

Problem 9 (Sp12, Sp11). Suppose the power series $\sum_n a_n x^n$ converges for all real x , and the smooth real valued function f has the property that

$$\lim_{x \rightarrow 0} \frac{f(x) - \sum_{j=0}^n a_j x^j}{x^n} = 0$$

for all n . Prove or give a counterexample to the claim that $f(x) = \sum_n a_n x^n$.

Problem 10 (Sp12). Find the Fourier series of the function with period 2π that is 1 if $|x| < \varepsilon$ and 0 if $\varepsilon \leq |x| < \pi$. Find the sum

$$\frac{\sin 1}{1} + \frac{\sin 2}{2} + \frac{\sin 3}{3} + \dots$$

Problem 11 (Sp12, Sp16, Sp17). Find the structure of the group of units of $\mathbb{Z}_{2012}$.

Problem 12 (Fa99, Sp12). Evaluate the integral

$$I = \frac{1}{2\pi i} \int_{|z|=1} \frac{(z+2)^2}{z^2(2z-1)} dz$$

where the direction of integration is counterclockwise.

Problem 13 (Sp12). How many conjugacy classes of nilpotent 5×5 complex matrices are there, up to conjugacy by invertible matrices?

Problem 14 (Fa78, Sp12). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $f(x) \leq f(y)$ for $x \leq y$. Prove that the set where f is not continuous is finite or countably infinite.

Problem 15 (Sp12). Suppose $f(x) \in \mathbb{Q}[x]$ is a polynomial with rational coefficients, n is a positive integer, $nf(x) \in \mathbb{Z}[x]$ has integer coefficients and $f(m)$ is an integer for all integers m , $0 \leq m < n$. Show that $f(m)$ is an integer for all integers m .

Problem 16 (Sp12). Let $f, g : \mathbb{C} \rightarrow \mathbb{C}$ be entire functions. Assume $\Re f(z) \geq \Re g(z)$ for all z such that $|z| = 1$. Show $\Re f(z) \geq \Re g(z)$ for all z such that $|z| < 1$.

Problem 17 (Sp05, Sp12). Suppose that V is a finite dimensional vector space over a field $\mathbb{F}$, and A, B are commuting diagonalizable linear maps from V to V . Show AB is diagonalizable.

Problem 18 (Sp12). Show that there are infinitely many integer solutions of $x^2 - 2y^2 = 7$.